

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 5 — Normalization — 2NF and 3NF
Bookstore & Hospital Management Systems

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Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	05
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Continue the bookstore decomposition from 1NF through 2NF to 3NF.
- Remove partial dependencies (2NF) and transitive dependencies (3NF).
- Independently design and normalize a Hospital Management System from scratch.

2. Database Structure

Two independent databases are used to demonstrate normalization end-to-end.

2.1 bookstore_norm (2NF)

Table	Purpose
Order_2NF	Order header, separated from item-level details
Book_2NF	Book details, separated so they aren't repeated per order line
OrderDetail_2NF	Order line items, linking orders to books

2.2 bookstore_norm (3NF)

Table	Purpose
Customer_3NF	Customer details, separated out of the order record
Book_3NF	Book details (3NF-clean)
Order_3NF	Order header referencing a customer
OrderDetail_3NF	Order line items (3NF-clean)

2.3 hospital_norm

Table	Purpose
Hospital_1NF	Starting unnormalized hospital record
Departments	Hospital departments
Doctors	Doctors, linked to a department
Patients	Patient records
Visits	Links patients to doctors for each visit

3. Key Tasks / Concepts Implemented

- Identifying partial dependencies in the 1NF bookstore table and decomposing into 2NF tables.
- Identifying transitive dependencies in the 2NF result and decomposing further into 3NF tables.

- Designing a fresh Hospital Management System schema (Departments, Doctors, Patients, Visits) normalized directly to a clean relational form.
- Inserting sample data and verifying relationships with SELECT/JOIN queries.

4. Implementation (phpMyAdmin)

Both databases were created, populated, and verified using phpMyAdmin under XAMPP.

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ book_2nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orderdetail_2nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_2nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
3 tables	Sum	0	InnoDB	utf8mb4_general_ci	64.0 KiB	0 B

☐ Check all
 With selected:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ book_2nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ book_3nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ customer_3nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ orderdetail_2nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ orderdetail_3nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ order_2nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ order_3nf	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
7 tables	Sum	0	InnoDB	utf8mb4_general_ci	160.0 KiB	0 B

☐ Check all
 With selected:

Structure tabs of 2NF and 3NF tables

Server: 127.0.0.1 » Database: hospital_norm

Structure SQL Search Query Export Import Operations Privileges Routines Events More

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ departments		2	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ doctors		2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ hospital_1nf		4	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ patients		3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ visits		4	InnoDB	utf8mb4_general_ci	48.0 KiB	-
5 tables	Sum	15	InnoDB	utf8mb4_general_ci	128.0 KiB	0 B

☐ Check all With selected:

Structure tab of hospital_norm database

Server: 127.0.0.1 » Database: hospital_norm » Table: departments

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

`SELECT * FROM `departments``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Extra options

	Specialty	DeptName	DeptHead
☐	Cardiology	Heart Care	Dr. Tariq
☐	Dermatology	Skin Clinic	Dr. Asma

☐ Check all With selected:

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Server: 127.0.0.1 » Database: bookstore_norm » Table: customer_3nf

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More

Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

`SELECT * FROM `customer_3nf``

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		CustID	CustName	CustEmail
☐	Edit Copy Delete	C-11	Bilal	bilal@x.com
☐	Edit Copy Delete	C-12	Areeba	areeba@x.com

↑ ☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Browse tabs for both databases (to show sample data)

5. Conclusion

This lab completed the normalization journey started in Lab 4, taking the bookstore schema to full 3NF, and independently proved the same skill by designing a normalized Hospital Management System from a blank slate.